3 Thermodynamic potentials

The problems are roughly ordered to coincide with the lectures. Those indicated with ♡ are especially important. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

Problem 3.1 **Rubber: a more complicated, and interesting, internal energy**
A piece of rubber with volume V and length L is subject to work by hydrostatic pressure, p, and by a tensional force f.

(i) Construct the expression for $\mathrm{d}U$.

(ii) Generate the potentials which have as natural variables (S, V, f), (S, p, f) and (T, p, f).

(iii) Derive the Maxwell relations involving the derivatives

$$\left(\frac{\partial T}{\partial V}\right)_{L,S}, \quad \left(\frac{\partial S}{\partial L}\right)_{T,p}, \quad \left(\frac{\partial S}{\partial f}\right)_{p,L}.$$

[Hint: the last one is a bit tricky. You have to consider the inverse derivative.]

(iv) ♣ Why is rubber (almost) uniquely an appropriate choice of material for this question?

Problem 3.2 **Gibbs-Helmholtz**
Show the Gibbs-Helmholtz equation

$$U = -T^2 \left(\frac{\partial}{\partial T}\right)_V \left(\frac{F}{T}\right),$$

where F is the Helmholtz free energy. *[Hint: start from the right-hand side.]*

Problem 3.3 **Equilibration problem**
An isolated system is composed of two compartments with volumes V_1 and V_2 and temperatures T_1 and T_2 containing gases in thermal contact. Write down an expression for the entropy differential of each gas. Maximising the total entropy of the system, find the equilibrium condition linking T_1 and T_2. *[Hint: the total energy is conserved and the volumes are constant.]*

Problem 3.4 **The World at constant pressure**

(i) Define the Gibbs free energy $G(T, p)$ for a gas. Show how it can be used in conjunction with the fundamental relation of thermodynamics to derive the Maxwell relation

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p.$$

(ii) Using part (i), show that for all gases, the heat capacity at constant pressure, $C_p(T,p)$, follows

$$\left(\frac{\partial C_p}{\partial p}\right)_T = -T\left(\frac{\partial^2 V}{\partial T^2}\right)_p \; .$$

(iii) The equation of state of a gas of hard spheres is $p(V-nb) = n\mathrm{R}T$, where n is the number of moles and b is a constant. Show that the heat capacity $C_p(T,p)$ of this gas is actually independent of p.

♣*Problem 3.5* **Hard labour**

A cylinder contains 0.1 kg of water at 15°C. A piston reversibly increases the pressure on the water isothermally from 1 atm to 100 atm. Find

(i) the work done on the water by the piston,

(ii) the heat removed from the water,

(iii) the change in the internal energy of the water.

For water at 15°C, the cubic expansivity, β, and the compressibility, κ, are respectively

$$\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p = 1.5\times 10^{-4}\ \ \mathrm{K}^{-1}\ , \quad \text{and} \quad \kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T = 4.9\times 10^{-12}\,\mathrm{Pa}^{-1}\ .$$

[Hint: the compressibility is so small that the volume will stay almost constant (water: $V = 1\,\mathrm{l}\,kg^{-1}$). For part (ii), the relation derived in question 4 part (i) will help.]
[$W = 25\,\mathrm{mJ},\ Q = 44\,\mathrm{J}, \Delta U = -44\,\mathrm{J}$.*]*

(iv) ♣ What would be the change in temperature of the water if the increase in pressure were made adiabatic? *[Hint: you can suppose that the specific heat capacity of water is $C_p = 4.2\,\mathrm{kJ\,kg^{-1}\,K^{-1}}$ at all pressures.]*[$\Delta T = 0.10\,\mathrm{K}$.]

♣*Problem 3.6* **Pressure squeezed from Clapeyron**

By integration of the Clapeyron equation, find $p(T)$ assuming that the molar latent heat L is a constant over the range of temperatures, that the specific volume of the vapour is much larger than the specific volume of the liquid, and that the vapour behaves like a perfect gas.